Sarah A. Carneiro, PhD.

AI Research Scientist |AI Research Engineer| Data Scientist
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Education

Ph.D. in Signal, Image, and Automation — Paris, France
Université Gustave Eiffel (UGE) — *2020 - 2024*

M.Sc. in Computer Science — Campinas, Brazil
University of Campinas (UNICAMP) — *2020*

Research Experience

Postdoctoral Researcher – Multimodal AI and Foundation Models

École Polytechnique, LIX, Paris, France — *2025–Present*

- Develop **multimodal foundation models** for scientific AI, integrating protein sequences, molecular structures, and natural language.
- Contribute to **Prot2Text-V2** (NeurIPS 2025), improving protein description generation by up to 4 points on standard text-generation metrics across a dataset of approximately 250,000 protein entries.
- Contribute to **PPI2Text** for protein–protein interaction captioning, achieving a reported BioBERT score of 86.33 using a dataset of 351,000 interaction pairs.
- Co-developed **CARTE**, a benchmark of French cultural and regional knowledge, evaluating 32 LLMs (1B–30B parameters) on 3,132 questions across 13 regions, with human-validated questions.
- Lead LLM benchmarking within **OPENLLM France**; conduct ablation studies and robustness evaluations.
- Contribute to collaborative research strategy and funding proposals for **Hi! PARIS** and **ANR**.

PhD Researcher – Structure-Aware Deep Learning for Spatio-Temporal Data

IFP Energies Nouvelles / Université Gustave Eiffel, France — *2020–2024*

- Developed a **topography-aware deep learning framework** for speed prediction in data-sparse regions using road clustering and sparse GPS trajectories.
- Achieved 78% lower MSE than an MLP baseline (6.75 vs. 31.03) using road-infrastructure features.

Research Scientist – Deep Learning for Spatio-Temporal Video Understanding

Semantix, Campinas, Brazil — *2017–2019*

- Developed multi-stream deep neural network ensembles for **human pose estimation and video understanding**, applying deep learning to visual representation learning and action analysis using AWS cloud computing infrastructure for model training and experiments.

Selected Publications

- **Prot2Text-V2: Multimodal Protein-to-Text Generation for Function Prediction.** NeurIPS, 2025.
- **PPI2Text: Captioning Protein-Protein Interactions with Coordinate-Aligned Pair-Map Decoding.** arXiv preprint, 2026.

- **CARTE: A Benchmark for Mapping Language Model Knowledge Across France.** arXiv preprint, 2026.
- **The Masked Advantage: Uncovering Local-Language Access to Cultural Knowledge in LLMs.** arXiv preprint, 2026.
- **Every Wrong Answer Counts: Option-Level Psychometrics for LLM Multiple-Choice Benchmarks.** arXiv preprint, 2026.
- **SWMLP: Shared Weight Multilayer Perceptron for Car Trajectory Speed Prediction Using Road Topographical Features.** MT-ITS, 2023.
- **Graph-Based Supervoxel Computation from Iterative Spanning Forest.** International Conference on Discrete Geometry and Mathematical Morphology, 2021.
- **Fight Detection in Video Sequences Based on Multi-Stream Convolutional Neural Networks.** SIBGRAPI, 2019.

Research Supervision and Mentoring

- **Foundation Models and LLMs:** Co-supervision of a research project on agentic systems and constrained language-model generation.
- **LLM Evaluation and Efficiency:** Supervision of a research project on benchmarking and evaluating large language models under quantization.
- **Medical Computer Vision:** Supervision of a research project applying computer vision methods to medical image analysis.
- **Synthetic Media Detection:** Supervision of an undergraduate research project on detecting AI-generated and manipulated images.
- **Applied Computer Vision:** Supervision of an undergraduate research project on automated plant disease classification.

Technical Skills

- **Multimodal AI and Foundation Models:** Multimodal representation learning, foundation models, cross-modal alignment and fusion, contrastive learning, biomolecular AI.
- **Deep Learning and Generative AI:** Large language models (LLMs), instruction tuning, LoRA, self-supervised learning, CNNs, GNNs, recurrent and temporal architectures.
- **Model Evaluation and Experimentation:** LLM benchmarking, ablation studies, reproducible evaluation, semantic and factuality metrics, data augmentation, robustness and generalization analysis.
- **GPU Computing and HPC:** CUDA-enabled PyTorch, multi-GPU experimentation, NVIDIA A100 GPUs, Linux, SSH, Slurm, batch-job submission and monitoring.
- **Programming and Frameworks:** Python, PyTorch, Hugging Face Transformers, TensorFlow, OpenCV, SciPy, Java, Git.
- **Languages:** Portuguese, English, French.

Certifications

- AI Literacy for Business Leaders — IBM, 2026 — AI strategy, business value assessment, and governance.
- Generative AI with Large Language Models — AWS, 2024 — LLM training, fine-tuning, evaluation, and application development.

Academic Service and Research Outreach

- **Symposium on Generative AI for Biochemistry and Health:** Member of the scientific committee, contributing to event organization, outreach, and invitations to academic and industry speakers.
- **Peer Review:** NeurIPS, ICLR, ACL.